

Yi-Ting Lai

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EDUCATION

National Yang Ming Chiao Tung University

Hsinchu, ROC

MS Computer Science, *Advisor: Yi-Ping You* GPA: 3.92/4.3

Sep 2023 – Present

- Thesis (tentative): “*AutoTex: Reducing Uncoalesced Global Memory Traffic on GPUs Using Texture Memory*”
- Courses: Advanced Compiler, Physical Design Automation, Accelerator Architectures for Machine Learning

National Taipei University of Technology

Taipei, ROC

BS Electronic Engineering GPA: 3.96/4.0, **Rank: 1/116**

Jun 2023

- **1st Place**, In-Campus Calculus Competition (245 participants, 2019)

EXPERIENCE

Advanced Semiconductor Engineering, Inc.

Kaohsiung, ROC

IT Intern

Jul – Aug 2022

- Developed a defect detection pipeline using **YOLOv5**, contributing to automated yield analysis.
- Built data dashboards with **Apache Superset** to visualize real-time factory production and downtime metrics.

TEACHING ASSISTANTSHIPS

- NYCU EEDE30050 Parallel Programming Feb – Jun 2025
- NYCU CSIC30045 Parallel Programming Sep – Dec 2024
- NYCU CSIC30098 Compiler Design, **Excellent Teaching Assistant Award** Feb – Jun 2024
- NYCU CSCS10012 Introduction to Compiler Design Sep 2023 – Jan 2024
- NTUT 5901206 Object-Oriented Programming Feb – Jun 2021

PROJECTS

VitaminC | C++

2023 – Present

A simple C compiler, targeting **QBE** and **LLVM IR**.

- Implemented key C features, including functions, arrays, pointers, loops, conditionals, structs, and unions.

sT Compiler | C

2023

A compiler for sT, a simple Turing-complete language, targeting **JASM** to run on the **JVM**.

- Designed a context-free grammar with documented syntax and semantics, implemented using **Flex** and **Bison**.
- Structured into distinct **lexer**, **parser**, and **code generator** phases for clarity and maintainability.

FastShop | Python

2022 – 2023

A lightweight shopping website with **Flask** backed by **MariaDB**.

- Developed and deployed a **RESTful API** to ensure robust backend communication.
- Integrated **SQLAlchemy ORM** with MariaDB for efficient data handling.
- Achieved **99% test coverage** using unit tests within an automated **CI workflow**.

Concentration Judgment System | Python

2021 – 2022

A real-time student concentration analysis system for online courses.

- Developed a **PyQt5** GUI following the **MVC** pattern, with data visualization using **Matplotlib**.
- Integrated **Dlib** and **OpenCV** for face and posture detection, using **SciKit-Fuzzy** for judgment with fuzzy logic.

SKILLS

Programming: C/C++, CUDA, Python, Go, Rust, Java, Bash

Software & Tools: Linux, Git, Make/CMake, GDB, Docker, LLVM, NVIDIA Nsight Compute/Systems, perf

Language: Mandarin Chinese (Native), English (Professional Proficiency, TOEIC 875), Japanese (Basic)

PUBLICATIONS

1. Yi-Ting Lai, Meng-Shiuan Pan. “A Concentration Judgment System for Online Courses: Design and Implementation.” VTS Asia Pacific Wireless Communications Symposium (APWCS), 2023.